

PCIe has undergone various improvements across a few generations, the above table lists the bandwidth for PCIe 3.0. PCIe 4.0 which is due to come in the next two years will effectively double the bandwidth shown above.

## 9. Bells and whistles

Each manufacturer adds a little something to their own motherboard to set it apart from the competition. Some manufacturers use more copper in the PCB thus making it into one large heatsink and thus all components are much cooler. Certain manufacturers use military grade components which are rated for higher temperatures and are more durable.

Then there are enthusiast boards which are geared towards overclocking, each manufacturer has their own line up. These boards have really high end onboard sound with isolated power supply. They have customised BIOS which can be remotely monitored and settings can be changed on the fly without the need for a system restart. Other components that are often customised include better capacitors, better VRM circuitry, Gigabit ethernet port etc.

## What to buy

When deciding on the motherboard keep the following points in mind

1. This is the first component that you are going to buy so list out all the features that you want so that you can select the chipset needed for your computer. Then list out what all connectivity options you need and raise that number by 1 or 2 for future upgrades.
2. Certain motherboards will have a physical 16x PCIe connector but internally they'll be wired for just 8x bandwidth. Now this is absolutely fine for SLI / CrossFire configurations on both PCIe 2.0 and PCIe 3.0 ports. But when using a single graphics card then a 16x port is necessary.
3. A power phase design with 12 phases is more than enough for moderate overclocking. Most records are set on mid-range boards with a few physical modifications made by the overclocker.
4. Opting for features and then never using them is a waste of money, so unless you are absolutely sure that you are going to buy that extra add-on card within the next year or so, don't go for the extra slots.
5. The placement of SATA ports right below the x16 PCIe slot may cause them to be inaccessible after a graphics card has been inserted, so take that into consideration.